



Linear Transformations · Mark Scheme

Core Pure 1 Chapter 7: Linear Transformations. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – Standard matrices

- 1 $(4, -11)$ B1; $(-1, 5)$ B1.
- 2 (a) Rotation 90° anticlockwise about O B1. (b) Reflection in the y -axis B1.
- 3 (a) $\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$ B1 (b) $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$ B1 (c) $\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ B1.
- 4 (a) The columns are the images: $\begin{pmatrix} 3 & -2 \\ 1 & 4 \end{pmatrix}$ B1. (b) $3x - 2y = 4$, $x + 4y = 6$, or use the inverse matrix $\frac{1}{14} \begin{pmatrix} 4 & 2 \\ -1 & 3 \end{pmatrix}$ M1; solving M1; $(2, 1)$ A1.
- 5(a) Rotation B1; 60° anticlockwise about O B1.
- 5(b) Six rotations of 60° make 360° , which returns every point to itself B1.
- 5(c) Rotation about the x -axis B1; through 90° (anticlockwise, looking from the positive x -axis towards O) B1.
- 5(d) $(2, 1, 3)$ B1.
- 5(e) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$ B1.

Section B – Combining transformations

- 6 QP M1; $= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ A1.
- 7 PQ M1; $= \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$ A1; reflection in the line $y = -x$ B1.
- 8 $\det \mathbf{M} = -2 - 5 = -7$ B1; area $= 7 \times 6 = 42 \text{ cm}^2$ M1; negative determinant: the orientation is reversed (the transformation includes a reflection) A1.
- 9 $\det \mathbf{M} = k^2 - 4$ B1; area scale factor $|k^2 - 4| = 5$ M1; $k^2 - 4 = 5 \Rightarrow k = \pm 3$ A1; $k^2 - 4 = -5$ gives $k^2 = -1$, no real solutions A1.
- 10(a) $\mathbf{U} = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$ B1; $\mathbf{V} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ B1.
- 10(b) $\mathbf{W} = \mathbf{VU}$ M1; $= \begin{pmatrix} 0 & 1 \\ -3 & 0 \end{pmatrix}$ A1.
- 10(c) Original area $= \frac{1}{2} \times 2 \times 3 = 3$ and $\det \mathbf{W} = 3$ M1; image area $= 9$ A1.
- 10(d) $\mathbf{W}^{-1} = \begin{pmatrix} 0 & -\frac{1}{3} \\ 1 & 0 \end{pmatrix}$ B1; $= \mathbf{U}^{-1}\mathbf{V}^{-1}$: a rotation of 90° anticlockwise about O , followed by a stretch parallel to the x -axis with scale factor $\frac{1}{3}$ B1.

Section C – Invariant points and lines

- 11 $\begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 4 \\ 2 \end{pmatrix} = \begin{pmatrix} 12 - 8 \\ 4 - 2 \end{pmatrix}$ M1; $= \begin{pmatrix} 4 \\ 2 \end{pmatrix}$, so the point is invariant A1.
- 12 $3x - 4y = x$ and $x - y = y$ M1; both give $x = 2y$ A1; $y = \frac{1}{2}x$ A1.



13 $\begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} t \\ 2t \end{pmatrix} = \begin{pmatrix} 3t \\ 6t \end{pmatrix}$ *M1*; $6t = 2(3t)$, so the image lies on $y = 2x$ *A1*; e.g. $(1, 2) \mapsto (3, 6) \neq (1, 2)$, so points are not individually fixed *B1*.

14 $\begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} x \\ mx \end{pmatrix} = \begin{pmatrix} (4+m)x \\ (2+3m)x \end{pmatrix}$ *M1*; $2+3m = m(4+m)$ *M1*; $m^2 + m - 2 = 0 \Rightarrow m = 1$ or $m = -2$ *A1*; $y = x$ and $y = -2x$ *A1*.

15(a) $\det \mathbf{M} = -\frac{9}{25} - \frac{16}{25} = -1$ *B1*; areas are unchanged and the orientation is reversed *B1*.

15(b) $\frac{3}{5}x + \frac{4}{5}y = x$ *M1*; $4y = 2x$ *A1*; $y = \frac{1}{2}x$ *A1*.

15(c) $\mathbf{M} \begin{pmatrix} t \\ -2t+c \end{pmatrix} = \begin{pmatrix} -t+\frac{4}{5}c \\ 2t-\frac{3}{5}c \end{pmatrix}$ *M1 A1*; $-2(-t+\frac{4}{5}c) + c = 2t - \frac{3}{5}c$, so the image lies on $y = -2x + c$ *A1*.

15(d) Reflection in the line $y = \frac{1}{2}x$ *B1*.

Roar-level challenge

16 Point $(x, mx+c)$ maps to $(x+3mx+3c, x-mx-c)$ *M1*; need $x-mx-c = m(x+3mx+3c)+c$ for all x *M1*; coefficients of x : $3m^2+2m-1=0$, so $m = \frac{1}{3}$ or $m = -1$ *A1*; constants: $-c = 3mc+c \Rightarrow c(3m+2) = 0$ *M1*; $3m+2 \neq 0$ for either value, so $c = 0$: the invariant lines are $y = \frac{1}{3}x$ and $y = -x$ (no vertical line is invariant) *A1*.